

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 - ASSESSMENT TOOL 2 – Industry Based Problems

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Industry Based problems	Assessment:	Assessment Tool 2 – Industry based
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyse and apply point estimation and confidence interval techniques to estimate population parameters and interpret statistical results in various industry-based problem-solving scenarios. (BL4)		
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Section / Batch:	AI&DS	Date:	19-08-2026

Code of Conduct

I Yedururi Anjani Priya (192324176) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Anjani Priya

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

		reasoning and insights				
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Online Retailer — Point Estimate of Proportion

Given:

- Sample size, ($n = 500$)
- Sample proportion, ($\hat{p} = 68\% = 0.68$)
- Population size = 50,000

The **point estimate of the population proportion** is the sample proportion:

$$\hat{p} = \frac{x}{n}$$

$$\hat{p} = 0.68$$

Answer:

$$\boxed{\hat{p} = 0.68 = 68\%}$$

Interpretation: We estimate that **68% of all 50,000 customers** made a purchase.

Estimated number of purchasing customers:

$$50000(0.68) = 34000$$

So, approximately **34,000 customers** are estimated to have made a purchase.

2. Manufacturing — Point Estimate of Population Mean

Given:

- Sample size = 100
- Sample mean = 1,250 hours

The point estimate of the population mean is the **sample mean**:

$$\begin{bmatrix} \hat{\mu} = \bar{x} \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mu} = 1250 \end{bmatrix}$$

Answer:

$$\begin{bmatrix} \boxed{\hat{\mu} = 1250 \text{ hours}} \end{bmatrix}$$

Interpretation: The estimated average lifetime of all electronic components is **1,250 hours**.

3. Banking — Point Estimates of Mean and Variance

Given:

- Sample size ($n=200$)
- Sample mean ($\bar{x}=\text{₹}18,500$)
- Sample standard deviation ($s=\text{₹}4,200$)

Point estimate of population mean

$$\begin{bmatrix} \hat{\mu} = \bar{x} \end{bmatrix}$$

$$\begin{bmatrix} \boxed{\hat{\mu} = \text{₹}18,500} \end{bmatrix}$$

Point estimate of population variance

The point estimate of population variance is the **sample variance**:

$$\hat{\sigma}^2 = s^2$$

$$s^2 = (4200)^2$$

$$s^2 = 17,640,000$$

Answers:

$$\boxed{\hat{\mu} = ₹18,500}$$

$$\boxed{\hat{\sigma}^2 = 17,640,000 \text{ ₹}^2}$$

4. Healthcare Analytics — Point Estimate of Proportion

Given:

- Total patients sampled = 400
- Patients experiencing side effect = 52

The point estimate of the population proportion is:

$$\hat{p} = \frac{x}{n}$$

$$\hat{p} = \frac{52}{400}$$

$$\hat{p} = 0.13$$

Convert to percentage:

$$[$$

$$0.13 \times 100 = 13\%$$

$$]$$

Answer:

$$[$$

$$\boxed{\hat{p} = 0.13 = 13\%}$$

$$]$$

Interpretation: We estimate that **13% of the entire patient population** would experience the particular side effect.

5. Food Industry — 95% Confidence Interval

Given:

- (n=64)
- Sample mean ($\bar{x}$ =498) g
- Sample standard deviation (s=12) g
- Confidence level = 95%

Since the sample size is (64), we can use the large-sample (z)-interval:

$$[$$

$$CI = \bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$]$$

For 95% confidence:

$$[$$

$$z_{\alpha/2} = 1.96$$

$$]$$

Step 1: Calculate Standard Error

$$[$$

$$SE = \frac{s}{\sqrt{n}}$$

$$]$$

$$[$$

$$SE = \frac{12}{\sqrt{64}}$$

$$]$$

$$[$$

$$SE = \frac{12}{8} = 1.5$$

]

Step 2: Calculate Margin of Error

[

$$ME = 1.96(1.5)$$

]

[

$$ME = 2.94$$

]

Step 3: Calculate Confidence Interval

[

$$498 \pm 2.94$$

]

Lower limit:

[

$$498 - 2.94 = 495.06$$

]

Upper limit:

[

$$498 + 2.94 = 500.94$$

]

Final Answer

[

$$\boxed{495.06 \text{ g} < \mu < 500.94 \text{ g}}$$

]

Interpretation: We are **95% confident** that the true average weight of all packets lies between **495.06 g** and **500.94 g**.

Final Answers at a Glance

No.	Problem	Final Answer
1	Population proportion	68%
2	Population mean	1,250 hours
3	Population mean	₹18,500
3	Population variance	17,640,000 ₹²

No.	Problem	Final Answer
4	Population proportion	13%
5	95% Confidence Interval	(495.06 g, 500.94 g)